

Copula Entropy

Theory and Applications

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Lecture 2: Theory of Copula Entropy (2)

① Physical Meaning

② Estimation

③ Growth of Theory

Physical Meaning of Copula Entropy

- **Laws of Thermodynamics**

- First law: Conservation of energy

Energy cannot be created or destroyed and only changes from one form to another.

$$\Delta U = Q - W \quad (1)$$

- Second law: Entropy increase

The total disorder of an isolated system always goes up over time.

$$\delta Q = TdS \quad (2)$$

- Third law: Absolute zero limit

As temperature approaches absolute zero (0 K), a system's entropy approaches a constant minimum.

- **Physical meaning of copula entropy**
entropy reduction due to increasing order of thermodynamic systems

$$H_c(\mathbf{x}) \leq 0 \quad (3)$$

Physical Meaning of Copula Entropy

- **Systems that keep order**

- Earth
constantly receiving energy from Sun to keep earth system evolving
- Society
receiving energy from natural environment to keep society running
- Life
getting food and energy from outside to keep alive

Copula Entropy: Estimation

- **Parametric** Estimation by Definition

$$H_c(\mathbf{x}) = -E(\log c(\mathbf{u}; \alpha)). \quad (4)$$

- ① estimating empirical copula density $\mathbf{u}$ based on rank statistic

$$u_i = F_i(x_i) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(x_t^i \leq x_i) \quad (5)$$

- ② estimating the parameter α of copula density function $c(\cdot)$ with maximum likelihood method
- ③ calculating according to (4)

Copula Entropy: Estimation

• Non-Parametric Estimation Method¹

- ① estimating empirical copula density u based on rank statistic

$$u_i = F_i(x_i) = \frac{1}{T} \sum_{t=1}^T \mathbf{1}(x_t^i \leq x_i) \quad (6)$$

- ② estimating copula entropy with the kNN entropy estimation method

• Advantages

- distribution-free, non-parametric
- tuning-free, insensitive to parameters
- good convergence
- easy to implement
- low computation burden

¹Jian Ma and Zengqi Sun. "Mutual Information Is Copula Entropy". In: *Tsinghua Science & Technology* 16.1 (2011). See also arXiv preprint arXiv:0808.0845, 2008, pp. 51–54.

Copula Entropy: Estimation

- **Conditional Mutual Information** Estimation

can be done with copula entropy estimators according to (7).

$$I(x; y|z) = H_c(x, z) + H_c(y, z) - H_c(x, y, z). \quad (7)$$

- 1 estimating the CE terms in (7) with CE estimator
- 2 calculating the result

Growth of Theory

Classical Information Theory

- CE based MI estimation, such as GCMI², MICE³, Vector Copula based MI estimation⁴, Vine copula based MI and CMI estimation⁵, semi-parametric MI estimation⁶, kernel based CE estimation⁷, non-parametric CE estimation⁸;

²Robin A.A. Ince et al. "A statistical framework for neuroimaging data analysis based on mutual information estimated via a gaussian copula". In: *Human Brain Mapping* 38.3 (2017), pp. 1541–1573.

³Adrien Cortes. "Accélération des techniques d'analyse de facteurs par méthodes d'apprentissage statistique et application à un GSE". Mémoire. Institut des Sciences Financières et d'Assurance, 2023.

⁴Yanzhi Chen et al. "Neural Mutual Information Estimation with Vector Copulas". In: *Advances in Neural Information Processing Systems*. 2025.

⁵王洪安. "上肢运动的肌间耦合分析方法研究". 硕士学位论文. 杭州电子科技大学, 2021, Houman Safaai. "Amortized Vine Copulas for High-Dimensional Density and Information Estimation". In: *arXiv preprint arXiv:2604.20568* (2026). arXiv: 2604.20568 [cs.LG]. URL: <https://arxiv.org/abs/2604.20568>.

⁶M. Mohammadi, M. Hashempour, and M. Emadi. "Semiparametric estimation of mutual information for elliptical copulas". In: *The 7th Seminar on Copula Theory and its Applications*. 2023, p. 44.

⁷林哲緯. "基於 Copula Entropy 與核密度估計的變數選取方法". 硕士学位论文. 国立政治大学, 2025.

⁸Baby A.K. and Rajesh G. "Nonparametric Estimation of Copula Entropy for Left Truncated Right Censored Data". In: *International Journal of Statistics and Reliability Engineering* 13.1 (2026), pp. 10–17.

Growth of Theory

Classical Information Theory

- asymmetric MI estimation⁹, CE^{210} , Bayesian dependence measure¹¹, and tail dependence measure¹²;

⁹Soumik Purkayastha and Peter X.K. Song. "Asymmetric predictability in causal discovery: an information theoretic approach". In: *arXiv preprint arXiv:2210.14455* (2022).

¹⁰Lin Liu, Cong Hu, and Xiao-Jun Wu. " CE^2 : A Copula Entropic Mutual Information Estimator for Enhancing Adversarial Robustness". In: *Pattern Recognition and Computer Vision*. Springer Singapore, 2024, pp. 163–174.

¹¹Guillaume Marrelec and Alain Giron. "An Inferential Measure of Dependence Between Two Systems Using Bayesian Model Comparison". In: *IEEE Transactions on Systems, Man, and Cybernetics: Systems* 55.3 (2025), pp. 1671–1683. DOI: 10.1109/TSMC.2024.3509025.

¹²Omid M. Ardakani and Rawan Ajina. "Tail risks in household finance". In: *Finance Research Letters* 69 (2024), p. 106065. ISSN: 1544-6123. DOI: <https://doi.org/10.1016/j.frl.2024.106065>. URL: <https://www.sciencedirect.com/science/article/pii/S154461232401095X>.

Growth of Theory

Classical Information Theory

- CE based directed information decomposition¹³;

$$DI(X^n \rightarrow Y^n) = \sum_{i=1}^n [I(X^i, Y^i) - I(X^i, Y^{i-1})] - I(Y^n) \quad (8)$$

¹³Aleksander Wieczorek and Volker Roth. "Causal Compression". In: *arXiv preprint arXiv:1611.00261* (2016),
Dinu Johannes Kaufmann. "Semi-parametric Gaussian Copula Models for Machine Learning". PhD thesis. University of Basel, 2017.

Growth of Theory

Classical Information Theory

- CE based maximal entropy¹⁴ and maximal Tsallis CE¹⁵.

¹⁴進藤晋. “Copula Entropy の最大化”. In: 日本 OR 学会 2011 年度秋季研究発表会. 2011, pp. 362–363, Milan Bubák and Mirko Navara. “Fitting Copulas with Maximal Entropy”. In: *Entropy* 27.1 (2025), p. 87. ISSN: 1099-4300. DOI: 10.3390/e27010087. URL: <https://www.mdpi.com/1099-4300/27/1/87>, Seyedeh Azadeh Fallah Mortezaejad et al. “Non-Parametric Multivariate Control Chart Using Copula Entropy”. In: *Sankhya B* 87 (2025), pp. 468–518. URL: <https://doi.org/10.1007/s13571-025-00374-y>.

¹⁵Seyedeh Azadeh Fallah Mortezaejad, Gholamreza Mohtashami Borzadaran, and Bahram sadeghpour Gildeh. “Joint dependence distribution of data set using optimizing Tsallis copula entropy”. In: *Physica A: Statistical Mechanics and its Applications* 533 (2019), p. 121897. ISSN: 0378-4371. DOI: <https://doi.org/10.1016/j.physa.2019.121897>. URL: <https://www.sciencedirect.com/science/article/pii/S0378437119311161>.

Growth of Theory

Graph Theory

- CE based graph matching¹⁶.

¹⁶Francisco Escolano et al. "The mutual information between graphs". In: *Pattern Recognition Letters* 87 (2017), pp. 12–19. DOI: <https://doi.org/10.1016/j.patrec.2016.07.012>. URL: <https://www.sciencedirect.com/science/article/pii/S016786551630174X>, Manuel Curado Navarro. "Structural Similarity: Applications to Object Recognition and Clustering". PhD thesis. Universitat d'Alacant, 2018.

Growth of Theory

Information Bottleneck

- CE based information bottleneck computation¹⁷.

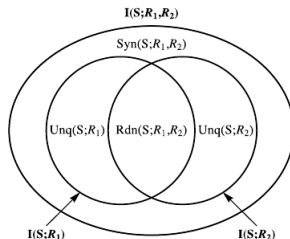
$$\mathcal{L}(\tilde{x}; x, y) = I(\tilde{X}; X) - \beta I(\tilde{X}; Y) \quad (9)$$

¹⁷Mélanie Rey and Volker Roth. "Meta-Gaussian information bottleneck". In: *Advances in Neural Information Processing Systems 25* (2012), Mélanie Rey. "Copula Models in Machine Learning". PhD thesis. University of Basel, 2015, Mario Wieser et al. "Inverse Learning of Symmetries". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020, pp. 18004–18015, Mario Wieser et al. "Learning Sparse Latent Representations with the Deep Copula Information Bottleneck". In: *International Conference on Learning Representations*. 2018. URL: <https://openreview.net/forum?id=Hk0wHx-RW>, Mario Wieser. "Learning Invariant Representations for Deep Latent Variable Models". PhD thesis. University of Basel, 2020.

Growth of Theory

Partial Information Decomposition

- unique information estimation¹⁸, synergy estimation¹⁹, and Ω information estimation²⁰.



¹⁸Ari Pakman et al. "Estimating the Unique Information of Continuous Variables in Recurrent Networks". In: *Advances in Neural Information Processing Systems* 34 (2021), pp. 20295–20307.

¹⁹Vlad-Bogdan Coroian, Pedro Mediano, and Tolga Birdal. *Scaling up synergy estimators*. Tech. rep. Imperial College London, 2024.

²⁰Laouen Belloli and Rubén Herzog. *THOI: An efficient library for higher order interactions analysis based on Gaussian copulas enhanced by batch-processing*. GitHub. URL: <https://github.com/Laouen/THOI>. 2024. URL: <https://pypi.org/project/thoi/>.

Growth of Theory

Information Geometry

- copula variational Bayes algorithm²¹.

$$D_{KL}(\tilde{p}(\mathbf{x})|p(\mathbf{x})) = D_{KL}(\tilde{c}(\mathbf{u})|c(\mathbf{u})) + \sum_i D_{KL}(\tilde{p}_i(x_i)|p_i(x_i)) \quad (10)$$

$$\geq \sum_i D_{KL}(\tilde{p}_i(x_i)|p_i(x_i)) \quad (11)$$

²¹Robert Zimmerman. *Copulas and Information Geometry: Strange Bedfellows?* Tech. rep. URL:https://rob-zimmerman.ca/files/projects/STA4531_Copulas_IG_Strange_Bedfellows.pdf. University of Toronto, 2025.

Growth of Theory

Optimal Transport

- entropic optimal transport²².

$$D_{EOT}(\epsilon) = \min_{P(X,Y)} E[d(X, Y)] + \epsilon I(X, Y) \quad (12)$$

²²[Giuseppe Serra](#). "Goal-oriented Semantic Communication : A Rate Distortion Perception Approach". PhD thesis. Sorbonne Université, 2026.

Growth of Theory

Category Theory

- copula operad²³.

²³[Xuexing Lu](#). "Copula Operad and Copula Entropy". In: *arXiv preprint arXiv:2609.20512* (2026).

Growth of Theory

Copula Theory

- copula function estimation²⁴, and circular-linear copulas²⁵;

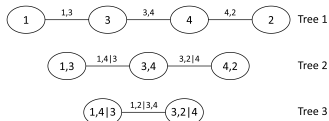
²⁴Longxia Qian et al. "A New Estimation Method for Copula Parameters for Multivariate Hydrological Frequency Analysis With Small Sample Sizes". In: *Water Resources Management* 36.4 (2022), pp. 1141–1157. ISSN: 1573-1650. URL: <https://doi.org/10.1007/s11269-021-03016-w>, Rafael Calsaverini and Renato Vicente. "An information-theoretic approach to statistical dependence: Copula information". In: *EPL (Europhysics Letters)* 88.6 (2009), p. 68003. DOI: 10.1209/0295-5075/88/68003. URL: <https://doi.org/10.1209/0295-5075/88/68003>, Rafael S. Calsaverini. "Tópicos em Mecânica Estatística de Sistemas Complexos". PhD thesis. Universidade de São Paulo, 2013.

²⁵Florian H. Hodel and John R. Fieberg. "cylcop: An R Package for Circular-Linear Copulae with Angular Symmetry". In: *bioRxiv* (2021), p. 2021.07.14.452253, Najibullah Karimi1 and Ali Dastbaravarde. "Circular-Linear Copulas: Properties and Applications". In: *15th Seminar on Probability and Stochastic Processes, University of Kurdistan*. 2025, pp. 200–211.

Growth of Theory

Copula Theory

- Vine copula model selection²⁶.



²⁶Fadhah Amer Alanazi. "Truncating Regular Vine Copula Based on Mutual Information: An Efficient Parsimonious Model for High-Dimensional Data". In: *Mathematical Problems in Engineering* 2021 (2021), p. 4347957. URL: <https://doi.org/10.1155/2021/4347957>, 王念鸽. "基于互信息的 Vine Copula 模型的高频数据投资组合风险测度研究". 硕士学位论文. 浙江财经大学, 2023, Jin Liu et al. "Vine-Copula Seismic Functionality Evaluation Method of Medical Room Systems Based on Shaking Table Tests". In: *Earthquake Engineering & Structural Dynamics* 54.6 (2025), pp. 1499–1519, Edith Kovács and Tamás Szántai. "Vine copulas as a mean for the construction of high dimensional probability distribution associated to a Markov Network". In: *4th Workshop on Vine Copula Distributions and Applications*. See also arXiv preprint arXiv:1105.1697. 2011, Dániel Pfeifer. "Matrix-based vine copula building algorithms". MA thesis. Budapest University of Technology and Economics, 2022, Houman Safaai and Alessandro Marin Vargas. "Dynamic Vine Copulas: Detecting and Quantifying Time-Varying Higher-Order Interactions". In: *arXiv preprint arXiv:2605.03061* (2026).

Growth of Theory

Causal Analysis

- causal compression²⁷, causal structure learning²⁸, LiNGAM²⁹, and time series causal network construction³⁰.

²⁷ Aleksander Wieczorek and Volker Roth. "Causal Compression". In: *arXiv preprint arXiv:1611.00261* (2016).

²⁸ Leonie Bossemeyer. "Machine Learning for Causal Discovery with Applications in Economics". MA thesis. Ludwig-Maximilians-Universität München, 2021, Marvin Lasserre, Régis Lebrun, and Pierre-Henri Wuillemin. "Learning Continuous High-Dimensional Models using Mutual Information and Copula Bayesian Networks". In: *Thirty-Fifth AAAI Conference on Artificial Intelligence, AAAI 2021*. AAAI Press, 2021, pp. 12139–12146. URL: <https://ojs.aaai.org/index.php/AAAI/article/view/17441>, Marvin Lasserre. "Apprentissages dans les réseaux bayésiens à base de copules non-paramétriques". PhD thesis. Sorbonne Université, 2022. URL: <https://tel.archives-ouvertes.fr/tel-03647090>.

²⁹ Joe Suzuki and Tian-Le Yang. "Generalization of LiNGAM that allows confounding". In: *2024 IEEE International Symposium on Information Theory (ISIT)*. IEEE, 2024, pp. 3540–3545, Hans Jarett J. Ong et al. "Redefining the Shortest Path Problem Formulation of the Linear Non-Gaussian Acyclic Model: Pairwise Likelihood Ratios, Prior Knowledge, and Path Enumeration". In: *arXiv preprint arXiv:2404.11922* (2025). URL: <https://arxiv.org/abs/2404.11922>.

³⁰ 陈琦琦. "基于 Copula 熵和偏秩相关系数的时序因果网络构建算法及其应用". 硕士学位论文. 合肥工业大学, 2023, Jing Yang and Xinzhi Rao. "Copula Entropy Based Causal Network Discovery from Non-stationary Time Series". In: *Pattern Recognition*. Springer Cham, 2025, pp. 115–131.

Growth of Theory

Machine Learning

- structure learning³¹, naive Bayes³², clustering³³, nonlinear PCA³⁴, decision trees³⁵, reinforcement learning³⁶.

³¹Luiz Desuó Neto et al. "Learning Continuous Decomposable Models Using Mutual Information and Statistical Copulas". In: *Entropy* 28.3 (2026), p. 293. ISSN: 1099-4300. DOI: 10.3390/e28030293. URL: <https://www.mdpi.com/1099-4300/28/3/293>, Luiz Desuó Neto. "Untying the Gordian knot of Bayesian inference in Markov networks". PhD thesis. Universidade de São Paulo, 2025.

³²Ábrahám Papp, Botond Szilágyi, and Edith Alice Kovács. "On Generalized Naive Bayes with Continuous Features". In: *arXiv preprint arXiv:2609.23819* (2026). arXiv: 2609.23819 [stat.ML]. URL: <https://arxiv.org/abs/2609.23819>.

³³刘磊 et al. "基于非线性相关性和复杂网络的径流相似性分区". In: *水科学进展* 33.3 (2022), pp. 442–451, Francesca Condino. "La divergenza di Jensen-Shannon nell'algoritmo di clustering dinamico per dati descritti da distribuzioni multivariate". PhD thesis. Università degli Studi di Napoli Federico II, 2009.

³⁴Yingpeng Wei and Li Wang. "Copula entropy-based PCA method and application in process monitoring". In: *2022 4th International Conference on Intelligent Information Processing (IIP)*. 2022, pp. 61–64. DOI: 10.1109/IIP57348.2022.00019.

³⁵Qingsong Shan and Qianing Liu. "Binary Trees for Dependence Structure". In: *IEEE Access* 8 (2020), pp. 150989–150998. DOI: 10.1109/ACCESS.2020.3017529, 罗良清 et al. "中国贫困治理经验总结：扶贫政策能够实现有效增收吗？". In: *管理世界* 38.2 (2022), pp. 70–83.

³⁶Yu-Xin Tian and Chuan Zhang. "A multimodal deep reinforcement learning framework for multi-period inventory decision-making under demand uncertainty". In: *Fuzzy Optimization and Decision Making* 24.4 (Dec. 2025), pp. 723–750. ISSN: 1573-2908. URL: <https://doi.org/10.1007/s10700-025-09462-0>.

Growth of Theory

Neural Networks and Deep Learning

- graph neural networks³⁷, IEGAIN³⁸, neural collapse³⁹, neural network pruning⁴⁰, and tree-copula variational autoencoders⁴¹.

³⁷ 陈佳雷 et al. “一种基于多要素注意力时空图卷积网络的径流预报方法”. Pat. CN117151285A. CN117151285A. 2023, 汤宇飞. “基于脉搏波的糖尿病和高血压诊断算法研究”. 硕士学位论文. 中国矿业大学, 2023.

³⁸ 武昊. “基于深度学习的化工过程软测量建模方法研究”. 博士学位论文. 北京化工大学, 2023.

³⁹ Siwei Wang and Stephanie E Palmer. “Towards understanding neural collapse in supervised contrastive learning with the information bottleneck method”. In: *arXiv preprint arXiv:2305.11957* (2024). URL: <https://arxiv.org/abs/2305.11957>.

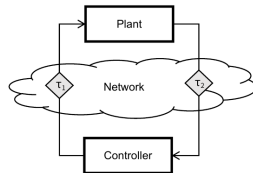
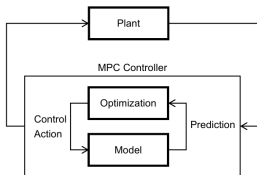
⁴⁰ Zihao Jiang et al. “CoPruning: Exploring the Parameter-Gradient Nonlinear Correlation for Neural Network Pruning Using Copula Function”. In: *OpenReview.net* (2024). URL: <https://openreview.net/forum?id=Yqqa9aNwB0>.

⁴¹ Yarden Rachamim and Shai Fine. “Beyond Mean-Field: Tree-Copula Variational Autoencoders for Structured Latent Dependencies”. In: *Proceedings of the 42nd Conference on Uncertainty in Artificial Intelligence*. 2026, pp. 5601–5622.

Growth of Theory

Control Theory

- model prediction controller⁴² and sliding mode controller⁴³.



⁴²Zhiwei Li et al. "Passenger-centric model predictive control for indoor temperature to facilitate energy-efficient airport terminal". In: *Building and Environment* 283 (2025), p. 113344. ISSN: 0360-1323. DOI: <https://doi.org/10.1016/j.buildenv.2025.113344>. URL: <https://www.sciencedirect.com/science/article/pii/S0360132325008200>.

⁴³Ting Yang et al. "Cooperative Voltage Control in Distribution Networks Considering Multiple Uncertainties in Communication". In: *Sustainable Energy, Grids and Networks* 39 (2024), p. 101459. DOI: [10.1016/j.segan.2024.101459](https://doi.org/10.1016/j.segan.2024.101459).

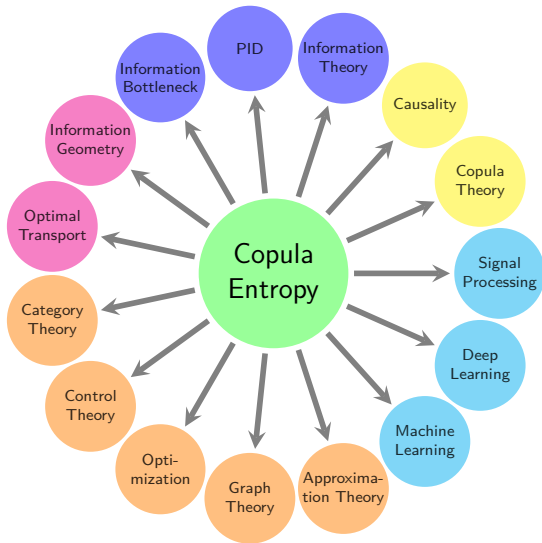
Growth of Theory

Signal Processing

- copula component analysis⁴⁸.

⁴⁸Jian Ma and Zengqi Sun. "Copula Component Analysis". In: *International Conference on Independent Component Analysis and Signal Separation*. 2007, pp. 73–80, Paola Palmatesta and Corrado Provasi. "Copula component analysis for dependence modelling". In: *Quaderni di Statistica* 14 (2012), pp. 185–188.

Growth of Theory



Monograph

Jian Ma. "Copula Entropy: Theory and Applications". In: *arXiv preprint arXiv:2025.18168* (2025)

Thanks!

Q&A